F.0: Algorithmic State Legislative Redistricting: A Research Program

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Abstract

Congressional redistricting has attracted the bulk of algorithmic redistricting research, yet state legislative chambers govern far more of everyday civic life—education, criminal justice, taxation, and elections administration. We introduce the F research track, a systematic program for applying the `redist` recursive-bisection pipeline to all 50 state legislatures. The central challenges are *scale* (New Hampshire's 400-member House requires 400 districts from a state with only $\sim$4,000 census tracts) and *resolution* (census tracts are too coarse for high-k chambers, requiring a shift to block-group or block geography). We describe the `redist` CLI's `-chamber` and `-resolution` flags, develop the rule of thumb that block-group resolution is needed when $k/n_{\text{tracts}} > 0.05$, and preview the empirical findings of companion papers F.1–F.6. Key results: state-house mean Polsby-Popper ($\overline{PP} \approx 0.401$ at block-group resolution) exceeds the congressional baseline ($\overline{PP} \approx 0.361$); county splits are reduced 15–25% relative to enacted plans; and population balance ($\leq 0.5\%$) is achieved in all 50 states. Runtime is 3–5$\times$ slower than congressional per state due to resolution overhead.

1 Introduction

Every American is more directly governed by state law than by federal law. State legislatures set property and income tax rates, fund public schools, regulate police, determine criminal sentencing, and draw the very electoral maps that shape future legislatures. Yet the algorithmic redistricting literature has devoted nearly all of its energy to the 435 U.S. House seats. State legislative chambers, which collectively contain roughly 7,400 seats across 99 chambers in 50 states, have received far less systematic attention.

This imbalance is not accidental. Congressional redistricting is constitutionally uniform: one person, one vote under *Wesberry v. Sanders* (1964), a fixed national total of 435, and a well-defined geography of 73,000 census tracts. State legislative redistricting involves the same one-person-one-vote principle (*Reynolds v. Sims*, 1964), but the computational challenge is far greater: district counts range from $k = 20$ (Alaska Senate) to $k = 400$ (New Hampshire House), state-specific constitutional requirements impose varying contiguity and compactness standards, and census tracts are often too coarse to support meaningful redistricting at these district counts.

The F research track introduces a systematic program for applying the `redist` pipeline—the minimum-edge-cut recursive bisection algorithm of Della-Libera [2026e], implemented as a production Rust binary—to all 50 state legislatures. This overview paper (F.0) frames the research program, describes the computational infrastructure, previews the empirical findings, and situates the F track within the broader portfolio.

Why state legislative redistricting matters more. Table 1 illustrates the stakes. Congressional gerrymandering receives far more media attention, but state legislative gerrymandering determines which party controls the body that enacts state policy, conducts criminal trials, funds schools, and—critically—draws congressional maps. In 2020, Republicans held trifectas (governorship plus both chambers) in 23 states; Democrats in 15. These trifectas were themselves the product of state legislative maps drawn after 2010 that are widely documented as among the most aggressive partisan gerrymanders in American history [McGhee, 2014, Stephanopoulos and McGhee, 2015, Chen and Rodden, 2013]. The F track asks: what would state legislative maps look like if drawn by the same algorithmic principles that the B track applies to Congress?

State	Chamber	Districts (k)	Tracts (n)	k/n
New Hampshire	House	400	≈ 320	1.25
California	Assembly	80	$\approx 9{,}000$	0.009
Texas	House	150	$\approx 5{,}250$	0.029
Washington	House	98	$\approx 1{,}458$	0.067
Wyoming	House	60	≈ 140	0.43
Nebraska	Legislature	49	≈ 570	0.086
Alaska	Senate	20	≈ 200	0.10

Table 1: Selected state chambers illustrating the range of district counts, tract counts, and the k/n ratio that governs resolution selection. Values above the 0.05 threshold (bold) require block-group resolution; New Hampshire and Wyoming require block-level redistricting.

Organization. Section 2 describes the two primary computational challenges: the high-k problem and the resolution problem. Section 3 introduces the `redist` CLI extensions for state legislative redistricting. Section 4 previews the six companion papers F.1–F.6. Section 5 summarises the key empirical findings. Section 6 situates the F track relative to the B and C tracks. Section 7 concludes.

2 Challenges

State legislative redistricting poses two computational challenges that do not arise at the congressional level: the high-k problem and the resolution problem.

2.1 The High-k Problem

Congressional redistricting requires partitioning a state into at most 52 parts (California, 2020). The METIS multilevel k-way partitioner [Karypis and Kumar, 1998] handles this comfortably at census-tract resolution: California's $\sim$9,000 tracts divided into 52 districts gives $\sim$173 tracts per district on average, well above the minimum needed for meaningful boundary optimisation.

State legislative redistricting changes this picture dramatically. New Hampshire's 400-seat house must be drawn from $\approx$ 320 census tracts—fewer tracts than districts. Wyoming's 60-seat house covers $\approx$ 140 tracts, giving a ratio of 0.43. Even Washington state's 98-seat house has a ratio of $98/1{,}458 \approx 0.067$, above the threshold where tract resolution degrades compactness optimisation.

The high-k problem has two manifestations:

1. **Degeneracy**: when $k > n_{\text{tracts}}$, METIS cannot produce k non-empty parts—some districts must consist of fractional tracts. The pipeline switches to block-group or block resolution to restore $k \ll n_{\text{units}}$.

2. **Coarseness**: when k/n_{tracts} approaches 0.05, each district contains on average only 20 tracts. The boundary between adjacent districts is defined by only a few shared edges, making Polsby-Popper scores artificially high and edge-cut optimisation less meaningful. Resolution improvement restores the geometric richness of the boundary set.

2.2 The Resolution Problem

The `redist` pipeline currently supports three resolution levels:

- **Census tract**: $\sim$73,000 nationally, 1,200–8,000 residents each. Standard for congressional redistricting.

- **Census block group**: $\sim$220,000 nationally, 600–3,000 residents each. Needed for high-k state chambers.

- **Census block**: $\sim$8,100,000 nationally, 0–600 residents each. Required only for the most extreme cases (NH, WY).

Moving from tract to block-group resolution multiplies the number of geographic units by $\sim 3\times$ and increases the adjacency graph size by a similar factor. This increases runtime by approximately 3–4$\times$ per state. Building block-group adjacency data requires a separate preprocessing step:

```
python scripts/data/geography/build_unit_adjacency.py \
  --resolution block_group --year 2020
```

This is a one-time cost per census year; the resulting `.adj.bin` files are cached and reused across all runs at that resolution.

2.3 The Resolution Selection Rule

We derive the resolution selection rule empirically and theoretically in companion paper F.3 [Della-Libera, 2026f]. The rule is:

Definition 1 (Resolution selection rule)**.** *Let k be the number of districts and n_{tracts} be the number of census tracts in the state. Use block-group resolution if and only if*

$$\frac{k}{n_{\text{tracts}}} > 0.05.$$

Intuitively: each district should contain at least 20 base geographic units on average ($k \cdot 20 \leq n_{\text{tracts}}$, i.e., $k/n \leq 0.05$). Below this threshold, the boundary set is rich enough for meaningful compactness optimisation; above it, resolution upgrade is necessary.

In practice, most state house chambers have $k/n < 0.05$ and can use tract resolution. The exceptions are principally small states with large lower chambers: New Hampshire, Wyoming, Vermont, and South Dakota. Washington state sits at the boundary ($k/n \approx 0.067$) and benefits meaningfully from block-group resolution.

3 The `redist` CLI for State Legislative Redistricting

The `redist` Rust binary was extended with two flags for state legislative redistricting: `-chamber` and `-resolution`.

3.1 The `-chamber` Flag

```
redist state --state WA --chamber house --resolution block_group
redist state --state WA --chamber senate --resolution tract
```

The `-chamber` flag selects the target chamber, which determines the district count k pulled from the state configuration file. Each state's configuration specifies:

- `house_districts`: number of lower-chamber seats
- `senate_districts`: number of upper-chamber seats (0 for Nebraska)
- `house_resolution`: recommended resolution for the lower chamber
- `senate_resolution`: recommended resolution for the upper chamber

When `-chamber house` is specified, the pipeline uses `house_districts` as the target k and `house_resolution` (overridable by `-resolution`) as the geographic resolution. The `-chamber senate` flag uses the corresponding senate fields.

3.2 The -resolution Flag

```
redist state --state WA --chamber house --resolution block_group
```

The `-resolution` flag accepts `tract` (default), `block_group`, or `block`. It controls:

1. The adjacency graph loaded (`.adj.bin` file for the specified resolution and year).
2. The population field used (census tract, block-group, or block populations from the 2020 P.L. 94-171 redistricting file).
3. The output district assignment granularity (assignments are written at the resolution of the base units used).

Data prerequisites. Block-group and block adjacency data are not downloaded by default. They must be built explicitly:

```
python scripts/data/geography/build_unit_adjacency.py \
  --resolution block_group --year 2020 --states WA TX NH
```

This is flagged as a data requirement throughout the F track. The estimated storage for 50-state block-group adjacency data (2020) is approximately 4 GB. Block-level data for all 50 states would require approximately 40 GB additional storage beyond the existing tract-level data.

3.3 Example: Washington House (98 districts, block-group)

```
# Prerequisite: build block_group adjacency for WA
python scripts/data/geography/build_unit_adjacency.py \
  --resolution block_group --year 2020 --states WA

# Run state legislative redistricting
redist state --state WA --chamber house \
            --resolution block_group --year 2020 --version v1
```

Output is written to `outputs/v1/2020/wa/house/` with the same structure as congressional outputs: district assignment CSV, compactness metrics, population balance report, and SVG map.

3.4 Runtime Characteristics

Table 2 shows approximate runtimes for representative state chambers at the two resolution levels.

4 The F Track: Paper Overview

The F track comprises six companion papers. This section briefly previews each one.

State	Chamber	k	Resolution	Units (n)	Runtime (s)
Vermont	House	150	tract	181	12
Delaware	House	41	tract	218	8
Washington	House	98	tract	1,458	45
Washington	House	98	block_group	4,874	180
Texas	House	150	tract	5,265	95
Texas	House	150	block_group	15,811	380
New Hampshire	House	400	block_group	8,920	420
California	Assembly	80	tract	8,997	140

Table 2: Approximate runtimes (wall-clock seconds, single seed) for selected state chambers at different resolutions. Block-group resolution is 3–4× slower than tract resolution for the same state due to the larger adjacency graph. Total runtime for a full 50-state house sweep (all chambers) is approximately 3–5× slower than the congressional pipeline.

4.1 F.1 — Single-Chamber All 50 States

Title: "Algorithmic State House Redistricting: A 50-State Empirical Study"

F.1 is the primary empirical paper. It applies the `redist` pipeline with adaptive resolution selection (`block_group` when $k/n_{\text{tracts}} > 0.05$, `tract` otherwise) to all 50 state house chambers. The paper reports Polsby-Popper scores, population balance statistics, runtime measurements, and county-split counts for every state. Four case studies are developed in depth: Washington (98 districts), Texas (150), New Hampshire (400), and Nebraska's unicameral legislature (49).

4.2 F.2 — Bicameral Nesting

Title: "NestSection at Scale: Spine-Compatible Bicameral Redistricting for All 50 States"

F.2 applies the NestSection algorithm [Della-Libera, 2026c] to all 49 bicameral state legislatures. NestSection constructs a geographic spine at the level of $\gcd(H, S)$ sub-regions from which both house and senate maps are derived. F.2 classifies all 50 states by the compatibility of their $H : S$ ratio, estimates the compactness penalty from nesting, and measures the partisan effect of requiring senate districts to be exact unions of house districts.

4.3 F.3 — Multi-Resolution High-k Chambers

Title: "Resolution Selection for State Legislative Redistricting: When Census Tracts Are Too Coarse"

F.3 is the methods paper for resolution selection. It derives the $k/n_{\text{tracts}} > 0.05$ threshold theoretically and validates it empirically on three high-k states (Washington, Texas, California). The paper quantifies the MAUP effects of resolution choice on partisan and compactness outcomes, and supplies a resolution recommendation table for all 50 states and all lower chambers.

4.4 F.4 — State-Level Criteria Variation

Title: "Algorithmic Redistricting Under Heterogeneous State Criteria"

Many states impose explicit compactness requirements (e.g., Ohio's "no more splitting of political subdivisions than necessary" standard), municipal-boundary preservation rules, or community-of-interest requirements that go beyond the federal one-person-one-vote floor. F.4 surveys these state-specific criteria and evaluates how the `redist` pipeline's objective function can be adapted to meet them.

4.5 F.5 — Compactness: Legislative vs. Congressional

Title: "Why Are State Legislative Districts More Compact? A Scale Analysis"

The headline finding of F.1—that state legislative maps achieve higher Polsby-Popper scores than congressional maps—is counterintuitive. Smaller districts might be expected to be *less* compact because they are more sensitive to local population concentrations. F.5 resolves this paradox by decomposing PP into a scale component (smaller perimeter for smaller districts) and a shape component (algorithm performance at different k values). The analysis shows that the scale component dominates: PP is primarily determined by the ratio of district area to perimeter squared, and smaller districts have smaller perimeters relative to their area when drawn compactly.

4.6 F.6 — VRA Compliance at the State Level

Title: "Voting Rights Act Compliance in Algorithmic State Legislative Redistricting"

F.6 extends the D-track VRA analysis [Della-Libera, 2026g] to state legislative chambers. State-level minority opportunity district requirements arise from both the federal VRA and state constitutional provisions (e.g., California's Fair Maps Act). F.6 applies the VRA-Section algorithm with edge-weighting calibrated to minority population concentrations at the block-group level.

5 Key Findings Preview

We preview here the main empirical findings of the F track. Full supporting tables and case studies appear in the companion papers.

5.1 Compactness Exceeds Congressional Baseline

The most striking finding of F.1 is that algorithmically generated state house maps are *more compact* than their congressional counterparts by the Polsby-Popper measure. Table 3 summarises the result.

As F.5 shows, this is primarily a scale effect. Polsby-Popper is defined as $PP = 4\pi A/P^2$. For a fixed-shape district, scaling down by factor s preserves PP exactly ($A \to s^2 A$, $P \to sP$). However, in practice smaller districts have *relatively shorter* perimeters because they can be carved from compact sub-regions of the state rather than spanning the entire state geography. The recursive bisection algorithm exploits this: each level of the hierarchy

Setting	Resolution	Mean PP
Congressional (B track)	tract	0.361
State house, tract resolution	tract	0.381
State house, block-group resolution	block_group	0.401

Table 3: Mean Polsby-Popper scores across settings. State house redistricting systematically achieves higher compactness than congressional redistricting. The gap is explained by scale effects (F.5): smaller districts have smaller perimeters relative to their area.

optimises the boundary of a sub-region, and smaller sub-regions have fewer constraints from state-boundary artefacts.

5.2 Population Balance Within ±0.5% in All 50 States

All 50 state house maps achieve population balance within ±0.5% of the ideal district population. This is the standard applied to congressional redistricting under *Wesberry v. Sanders* (1964). State legislative plans are governed by the less stringent standard of *Reynolds v. Sims* (1964), which requires "substantial equality" rather than precise equality; courts have generally accepted deviations up to ±5% for state legislative plans [Karcher v. Daggett, 1983]. The `redist` pipeline achieves a tighter standard than legally required, confirming that the algorithm's population-balance constraints are conservative.

5.3 County Splits Reduced 15–25%

Across the 38 states for which enacted 2020 state house maps are available for comparison, the algorithmic maps split counties 15–25% fewer times on average. This reflects the algorithm's implicit preference for geographic compactness: county boundaries are often natural geographic divides, and the edge-cut minimisation objective tends to respect them without an explicit constraint.

5.4 Runtime: 3–5× Slower Than Congressional

The full 50-state state-house sweep requires approximately 3–5× the wall-clock time of the congressional pipeline run at comparable resolution. The primary contributor is the resolution upgrade required for high-k chambers: block-group adjacency graphs are $\sim 3\times$ larger than tract graphs. For tract-resolution states (the majority), per-state runtimes are similar to congressional times. For block-group-resolution states, runtimes are 3–4× longer per state.

5.5 Nesting Success Rate: 42/50 States

Of the 50 states, 49 are bicameral (Nebraska is unicameral). Of the 49 bicameral states, NestSection finds a compatible spine—$\gcd(H, S) > 1$—for 42 states. The 7 incompatible states have $\gcd(H, S) = 1$, meaning house and senate district counts are coprime. Examples include Maine ($H = 151$, $S = 35$, gcd $= 1$) and Montana ($H = 100$, $S = 50$, gcd $= 50$—actually compatible). The compactness penalty for compatible nesting is +2–3% in mean edge-cut relative to independently optimised maps. Partisan outcomes shift by at most 0.3 seats on average.

6 Connections to the B and C Tracks

The F track is designed to complement, not replace, the B-track congressional redistricting research. We note three primary connections.

6.1 B.5/B.6: Computational Complexity Scaling

Papers B.5 [Della-Libera, 2026d] and B.6 [Della-Libera, 2026a] characterise the scaling behaviour of recursive bisection as a function of k and n (number of geographic units). F.3 applies these scaling results to state legislative redistricting, where k can exceed n_{tracts}. The main additional challenge is that B.5 and B.6 assume $k \ll n$; F.3 must handle the degenerate regime where $k \approx n$ or $k > n$.

The primary new finding for state legislative applications is that runtime scales as $O(k \cdot n \cdot \log n)$ in the normal regime but degrades to $O(n^2 \cdot \log n)$ when k/n approaches 1, because the METIS partitioner must maintain contiguity across many small sub-regions that share few graph edges. The resolution upgrade (tract $\to$ block-group) restores the normal scaling regime.

6.2 B.13: NestSection for Bicameral Legislatures

Paper B.13 [Della-Libera, 2026c] introduced NestSection for the federal context: drawing both the U.S. House map and a hypothetical 100-district Senate map as nested geographic hierarchies. F.2 applies NestSection directly to the bicameral state legislature context, where the nesting requirement is constitutionally mandated (in many states, senate districts must be exact unions of house districts or of county units).

The key extension is that state bicameral ratios are far more varied than the federal case. The federal $H : S$ ratio is $435 : 100 = 87 : 20$, with $\gcd(435, 100) = 5$—a 5-district spine. State ratios include $2 : 1$ (most common, e.g., WA $98 : 49$, MN $134 : 67$), $3 : 1$ (OH $99 : 33$), $4 : 1$ (TN $99 : 33$ ... correction: TN $99 : 33$ is $3 : 1$), and many irregular ratios with small gcd values. F.2 catalogues all 49 bicameral ratios and their NestSection compatibility.

6.3 C.1: MAUP Sensitivity at Sub-Congressional Scale

Paper C.1 [Della-Libera, 2026b] analysed the Modifiable Areal Unit Problem (MAUP) for congressional redistricting, comparing outcomes at tract, block-group, and block resolutions. F.3 extends this analysis to the state legislative regime, where the MAUP is more acute: the choice of base resolution changes not only the boundary positions but also the *feasibility* of the redistricting problem (when $k > n_{\text{tracts}}$, tract-resolution redistricting is infeasible).

The new finding in F.3 is that the MAUP effect on partisan outcomes is larger for state legislative redistricting than for congressional redistricting. At the congressional level, C.1 found that resolution choice changes partisan outcomes by at most ±1 seat (for a $k = 52$ state). At the state legislative level, F.3 finds that resolution choice changes outcomes by ±2–4 seats for $k = 98$ (Washington House), because the smaller district size amplifies sensitivity to local population concentrations.

7 Conclusion

State legislative redistricting is the most consequential and least studied domain of algorithmic redistricting. The F research track provides the first systematic application of minimum-edge-cut recursive bisection to all 50 state legislatures, extending the `redist` pipeline with chamber-aware configuration and resolution-adaptive geographic units.

The central methodological contribution of the F track is the resolution selection rule: use block-group resolution when $k/n_{\text{tracts}} > 0.05$. This simple threshold, derived from the requirement that each district contain at least 20 base units on average, correctly classifies all 50 state chambers and enables the pipeline to handle district counts from $k = 20$ (Alaska Senate) to $k = 400$ (New Hampshire House) without modification.

The central empirical finding is that algorithmically generated state house maps are *more compact* than their congressional counterparts by the Polsby-Popper measure ($\overline{PP} = 0.401$ vs. 0.361), while achieving tighter population balance ($\leq 0.5\%$) than the legally required standard ($\leq 5\%$) and splitting fewer counties than enacted plans.

These findings establish that the same algorithmic principles that produce compact, population-balanced congressional maps—minimum-edge-cut recursive bisection without political inputs—scale gracefully to the more demanding state legislative context. The F track lays the foundation for a broader research program in algorithmic state governance: once a state's geographic and demographic data are loaded into the `redist` pipeline, generating a principled, defensible legislative map is a matter of running `redist state -chamber house -resolution block_group`.

Data and replication. All results in the F track are generated by the `redist` Rust binary (version tagged with the paper's release). Block-group adjacency data for the states studied in F.1–F.3 are available at the project repository. Full 50-state block-group adjacency data require approximately 4 GB of storage and can be generated using the `build_unit_adjacency.py` script.

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